

AI in JobQuest

From an expert's knowledge to a scored playthrough

This section walks through the product flow from start to end and stops at each of the five places where AI works.

Each of your AI questions is answered at the step it belongs to:

- Is JobQuest AI-native, and what happens when the AI is down? (slides 2 and 11)
- How are generated scenarios checked, and how is a playthrough scored? (slides 4 to 7)
- Does the judging model need to be stronger than the generating model? (slide 8)
- Is there published evidence that an LLM judge is reliable? (slide 9)
- Which framework for the LLM judge? (slide 10)
- Comparison of technology solutions, including AI models (slide 12)

Status: this week produced the design decisions and the literature review. **No measurement has been run yet.** The first test is planned for week 42.

Five places where AI works

Step	AI does	A person decides
Authoring (no player is waiting): Domain Expert → AI-1 → raw knowledge → AI-2 with Content Admin → seed → AI-3 with Domain Expert → approved seed		
AI-1	Interviews the Domain Expert to collect what a seed needs	The expert confirms the record
AI-2	Turns the expert's input into a structured game seed	The expert sets every scoring key and rubric anchor, and approves the seed
AI-3	Runs test playthroughs and picks the cases the expert must see	The expert rules on them and approves publication
Play: approved seed → AI-4 → player answers → AI-5 → skill scores; flagged cases → AI-3 → Domain Expert		
AI-4	Writes connecting narration and follow-up questions during play	Nobody at run time; the engine and rules constrain it
AI-5	Scores the playthrough against the rubric	The expert reviews flagged cases

- Principle: AI drafts, connects, scores and filters. Domain Experts decide occupational content, scoring keys and publication.
- AI is present at every step of the content lifecycle and of play. Without it the core loop cannot run (definition on slide 11).

AI-1 interviews the Domain Expert

- **What it does.** An LLM asks and probes to collect what a seed needs: incidents, what a good and a poor response look like, facts of the situation. One interview per expert; the merged content marks where experts agree and where they differ.
- **How it is constrained.**
 - A human-written topic guide based on the critical incident technique (Flanagan, 1954) and on situational-judgment-test development practice (Whetzel et al., 2020). The model probes inside each topic only.
 - A tracker of seed fields still missing.
 - The expert is shown a summary and confirms it.
 - Every turn is stored verbatim, and each seed field links to the turn it came from.
- **Evidence that an LLM can hold an interview to a guide.** In 766 AI-led interviews, 94.8% to 99.8% of questions were open-ended, 95.7% to 99.6% neutral, and 0.1% to 0.4% hallucinated (Chopra & Haaland, working paper). A chatbot asking follow-up questions elicited more informative and specific answers than a static form, with about 600 participants (Xiao et al., 2020).

AI-2 drafts the game seed

- **What it does.** Helps the Content Administrator turn the experts' answers into a structured seed: context, characters, situations, canon events, and the scoring frame.
- **Rules.**
 - Drafts are written from the expert's own words.
 - **The Domain Expert sets every scoring key and every rubric anchor.** The model only drafts.
 - A seed passes automatic structure checks, then the expert's approval, before any player sees it.
- **Why experts own keys and rubrics.**
 - Rubrics generated by LLMs differ from the rubrics human graders use, and LLM graders take shortcuts; analytic rubrics that reflect human grading logic narrow the gap.
 - When GPT-4 drafted 210 examination questions, specialists required revision of 15% and judged 0.5% false (Klang et al., 2023).
- **Schema.** The seed follows evidence-centered design: what is measured, which behaviour is evidence for it, and which situation elicits that behaviour (Mislevy & Haertel, 2006; Shute, 2011).
- **Checkpoint 1 of 4 for generated content:** structure check by machine, content approved by the expert.

AI-3 helps the expert test

- **Problem.** A volunteer expert with about seven hours cannot read every free-text answer.
- **What it does.** Runs AI-4 and AI-5 on test playthroughs (experts, testers, simulated answers) and ranks the cases that need the expert:
 - cases where the three judges differ by more than one level;
 - generated turns that broke a run-time rule;
 - **plus a blind random sample of cases that were not flagged.**
- **The expert decides.** The expert can also refine a rubric anchor during review, so rubrics are versioned: people's criteria shift while they grade (Shankar et al., 2024).
- **Why the random sample is required.** Confidence signals from an LLM grader separate right from wrong grades only weakly (AUC 0.57 to 0.64; Ferrer et al., preprint), so flagged cases alone can hide errors. If n audited cases show no error, the error rate is at most about $3/n$ at 95% confidence (Hanley & Lippman-Hand, 1983).
- **Simulated players only find relative problems.** Persona-prompted LLMs answered with near-perfect, uniform accuracy, unlike 2,359 real students (Balog & Bakken, 2026); simulated profiles correlated with real students at $r = 0.72$ on multiple-choice items (Lu & Wang, 2024). Real test plays are needed too.
- **Test set for AI-4's run-time checks.** Scripted player turns in four classes (on-script, off-script, adversarial, rubric-fishing); violation rate per class. Not yet built.
- **Checkpoint 2 of 4.** These expert-scored test plays are also the validation data for the judge.
- **This phase.** No experts or real players yet: the team stands in for both and the advisor checks a sample, so results show the machinery works, not agreement with experts.

AI-4 narrates during play

- **The engine owns the plot.** The game engine, not the model, controls the order of canon events. Canon content is shown verbatim from the seed. A situation may have several approved canon variants chosen at random; the log records which variant was served.
- **What the model writes.** Only the connecting narration and the follow-up questions that fit the player's free-text answer. The whole seed is placed in the prompt; no retrieval and no fine-tuning. The rubric and scoring keys are not in its prompt.
- **Run-time checks are rules, not another model call,** so checking adds no waiting time: numbers and names must come from the seed or from the player's own text, and the engine counts follow-ups. A model from a different family reviews the logs afterwards.
- **Evidence.**
 - Holding authored structure outside free generation is the established approach in interactive narrative (Riedl & Bulitko, 2013).
 - When the source fits in the prompt, long-context prompting outperformed retrieval on average (Li et al., 2024).
- **Checkpoint 3 of 4** (rules at run time) **and 4 of 4** (log review afterwards).

AI-5 scores the playthrough

- **A separate judge.** The scoring model is not the narrating model (design constraint DC-01).
- **No reference answer.** Answers are free text; the judge scores each rubric criterion by how far the answer meets the behaviour described for each level.
- **One call per situation per judge,** with at most three criteria in a call. The call runs in the background when the situation ends; results appear on the summary screen.
- **Auditable output.** For each criterion: a verbatim quote from the player, then the rationale, then the level. The quote must be a substring of what the player typed.
- **A panel of three judges from three model families.** The median is the score. Cases where judges differ by more than one level go to the expert through AI-3. If one judge is unavailable, the system waits for all three.
- **Why not one call for everything.** Batching several rubric checks into one call trades accuracy for cost, and the loss is large when the evidence is spread over a long trajectory (Peng et al., 2026). A verdict on one rubric shifts depending on which other rubrics are judged with it (Yu et al., preprint).
- **Why a panel.** A five-judge vote matched the best single judge (Pearson correlation with humans 0.627 against 0.623; Kim et al., 2025), so a panel reaches the best judge's level without knowing in advance which judge that is. LLM judges also give different scores to the same item across runs (Haldar & Hockenmaier, 2025). We do not claim a panel is more accurate than a single judge.

Does the judge need to be stronger than the generator?

- **Answer: not relative to the generator.** The judge scores text written by a person, not the generator's output, so the two are not in competition.
- **What the judge does need.**
 1. **Enough capacity.** Small models aligned poorly with human raters (intraclass correlation below 0.20) and larger models moderately (0.41 to 0.57), against 0.45 between the two human raters (Jung et al., 2025).
 2. **Not necessarily a reasoning model.** Two reasoning models had lower validity than a smaller standard model in essay scoring (Yoshida, 2025).
- **Decision method.** Each judge family is tested at two tiers in week 42; the cheapest tier that is not clearly worse is chosen.

Is an LLM judge reliable?

What published evidence says

- **Answer: conditionally, not by default.**
- **Below human raters on average.** On a standard essay dataset, mean quadratic weighted kappa was 0.313 (ChatGPT) and 0.297 (Llama) against 0.817 for the best supervised system (Mansour et al., 2024).
- **Agreement nears human level** with a strong model, an explicit rubric and scored examples (report Section 2.5, [101] to [105]). This phase has the first two; examples come later, from expert rulings.
- **Weakest where it matters.** Automated scorers handle fully correct and fully incorrect answers well and are much worse on partially correct ones, while human agreement stays constant (Schleifer et al., 2026).
- **Only as good as its knowledge of the work.** Without a reference answer, LLM judges agreed with experts only on questions they could answer themselves (Krumdick et al., preprint); judging ability tracks problem-solving ability (Tan et al., 2025). JobQuest has no reference answer, so the judge must know IT work and the rubric must carry domain knowledge.
- **Biased toward its own family.** LLM judges favour their own generations (Panickssery et al., 2024) and models with mistakes like their own (Goel et al., 2025); hence a judge family other than the generator's.
- **Evidence is from another setting:** English essays and short answers, 2023–24 models; none on Vietnamese or role-play.
- **Consequence.** Reliability must be measured on our own task (research problem RP-2), with the criteria of Williamson et al. (2012): quadratic weighted kappa ≥ 0.70 , standardised mean difference ≤ 0.15 , and at most 0.10 below expert–expert agreement.

Framework for the LLM judge

- **Offline test harness: Inspect AI**, open source, maintained by the UK AI Security Institute. It provides repeated runs, several grader models in one run, a median reducer and an ordinal Krippendorff's alpha. Our rubric needs a custom scorer.
- **In the product:** our own TypeScript module with schema-validated output. Prompts and rubrics are kept in data files shared by both.
- These tools have no peer-reviewed papers. The cited science is in the method: rubric-based scoring with reasoning before the score (Liu et al., 2023) and panels of judges (Verga et al., preprint).

AI-native, and what happens when the AI is down

- **Definition adopted.** A system is AI-native when AI is constitutive of its core loop, so that removing the AI makes the central activity collapse or change fundamentally (Xu et al., preprint, a survey of 53 AI-native games), and when AI is part of design, operation and maintenance, not an added component (Ericsson, white paper). No standards body defines the term; ITU-R M.2160 uses it without a definition.
- **Applied to JobQuest.** AI works at all five steps. Without it the system cannot fit the narration to a free-text answer and cannot score that answer. There is no substitute path: no keyword scoring and no scripted replacement.
- **When the model provider fails.**
 - Generation and scoring, the core, stop.
 - The session pauses. Answers and progress are saved (NFR-04), and play resumes when the AI returns.
 - Pages that do not use AI (sign-in, occupation map, profile) stay available.
 - Switching to another provider is still AI, not a fallback (DC-02).
- **Why several providers.** A study of public incident reports found an incident on average every 5.6 days for the OpenAI API and every 5.2 days for the Anthropic API, a median recovery time of about one hour, and no correlation between providers (Chu et al., 2025).

Comparing technologies and AI models

- **Four model families:** one for generation, three for the judge panel. Judges differ from the generator (DC-01) and from each other; bias toward related models falls as the relation grows distant (Li et al., 2026).
- **Criteria:** Vietnamese; knowledge of IT work; response time; price; schema-constrained output; data terms.

Role	Candidates (USD / 1M tokens, in / out, 8–9 Oct)	SEA-HELM Vietnamese (18 Sep)
Judge, Anthropic	claude-sonnet-5-5 (2 / 10); claude-haiku-5-5 (0.10 / 0.50)	Sonnet 5: 77.0; Haiku 4.5: 67.7 (earlier models)
Judge, OpenAI	gpt-6.1-sol (2 / 10); gpt-6-luna (0.10 / 0.50)	GPT 5.6 Sol: 77.1; GPT 5.6 Luna: 71.9 (earlier models)
Judge, Google	gemini-3.8-flash (0.75 / 3.75); gemini-3.5-flash-lite (0.30 / 2.50)	78.4; Gemini 3.1 Flash Lite: 75.2
Narrator	deepseek-flash (0.30 / 1.20); qwen3.8-flash (0.15 / 0.47)	DeepSeek V4 Flash: 72.1; Qwen 3.8 Flash Next: 74.9

- **What public data cannot tell us:** IT knowledge for our three roles (no public benchmark); the newest small models (not on the Vietnamese leaderboard); response time measured from Vietnam.
- **Cost of one playthrough** (token counts assumed, narrator included): about 0.015 USD with small-tier judges, 0.086 with mid-tier. **Final choice after the week-42 test**, with agreement, response time and cost.

Waiting for game design

- The scoring scale: how many levels and what each level means. The scale $-1 / 0 / +1 / +2$ is **planned, not final**; the current sample scenarios describe three levels only.
- The competency scoring table for a playthrough, and the specific criteria given to the judge.
- The list of required facts in a seed, needed to measure fact preservation (NFR-15).
- How evidence from the play process (hints, time, follow-ups) counts, and how scores combine into a competency.
- The current sample scenarios and scoring are placeholders and will be revised.
- **What does not depend on game design:** the measure itself. Agreement between judge and expert can be computed on any ordinal scale.

A decision we need from you

A daily free quota with paid extra plays. One playthrough costs an estimated 400 to 2,200 Vietnamese dong in model calls. May we move from "free for everyone" to a number of free plays per day plus paid additional plays? We would implement only the daily quota in this phase.

Peer-reviewed

Balog & Bakken (2026). Findings of EMNLP 2026
Chu et al. (2025). ICPE 2025
Flanagan (1954). Psychological Bulletin, 51(4) [26]
Goel et al. (2025). ICML 2025
Haldar & Hockenmaier (2025). EMNLP 2025
Hanley & Lippman-Hand (1983). JAMA
Jung et al. (2025). NCME AIME 2025
Kim et al. (2025). BiGGen Bench. NAACL 2025
Klang et al. (2023). BMC Medical Education, 23
Li, D. et al. (2026). Preference Leakage. ICLR 2026
Li, Z. et al. (2024). EMNLP 2024 Industry Track
Liu et al. (2023). G-Eval. EMNLP 2023 [58]
Lu & Wang (2024). Generative Students. L@S 2024

Mansour et al. (2024). LREC-COLING 2024
Mislevy & Haertel (2006). EM:IP, 25(4)
Panickssery et al. (2024). NeurIPS 2024
Peng et al. (2026). EMNLP 2026
Riedl & Bulitko (2013). AI Magazine, 34(1)
Schleifer et al. (2026). BEA 2026
Shankar et al. (2024). UIST 2024
Shute (2011). Stealth assessment [20]
Tan et al. (2025). JudgeBench. ICLR 2025 [60]
Whetzel et al. (2020). PAD, 6(1) [27]
Williamson, Xi & Breyer (2012) [71]
Xiao et al. (2020). ACM TOCHI
Yoshida (2025). Findings of EMNLP 2025

Preprints, standards, vendor

Preprints and working papers

Chopra & Haaland. CESifo Working Paper 10666

Ferrer et al. (2026). arXiv:2603.29559

Krumdick et al. (2025). arXiv:2503.05061

Verga et al. (2024). arXiv:2404.18796 [74]

Xu et al. (2026). AI Native Games. arXiv:2607.00527

Yu et al. (2026). arXiv:2608.14684

Standards, vendor, documentation

ITU-R Recommendation M.2160-0 (2023)

Ericsson. Defining AI native (white paper)

UK AI Security Institute. Inspect AI documentation

AI Singapore. SEA-HELM, Vietnamese (18 Sep 2026)

Pricing pages: Anthropic, OpenAI, Google,
DeepSeek, Alibaba (8–9 Oct 2026)